

Olist Profitability & Customer Analysis

Executive Summary — Margin Drivers and Customer Retention Risk

This analysis reviewed order, product, payment, and delivery data (2016-2018) to identify where revenue is being converted into margin least efficiently, which customers represent the greatest retention risk relative to their value, and what predicts that risk. Three findings stand out as immediately actionable.

49.24% Telephony margin — weakest of any top-15 category	39.3% of revenue from customers at risk of churning	63.3% of revenue from just 10 of 71 categories	53% of churn model's signal driven by delivery delay
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Finding 1 — Margin Erosion in Telephony

Telephony generates R\$309,860 in revenue (14th among 71 categories) but converts only 49.24% of that into margin after shipping cost — nearly 21 points below the next-lowest top-15 category (housewares, 59.70%) and less than half the margin efficiency of the strongest performer (computers, 94.33%). Since shipping cost tracks fairly closely with revenue at the category level overall, this gap is unusually large and specific to telephony.

Recommended action:

- Review freight pricing and packaging weight/dimensions for telephony products specifically
- Compare telephony's shipping cost per order against similarly-priced categories to isolate the cause

Finding 2 — High-Value Customers Going Quiet

RFM segmentation identified a group of 22,676 customers (24.3% of the customer base) with historically high spend, contributing R\$6.05M — 39.3% of total revenue. Their average recency is 364 days: this segment has not purchased in nearly a year. Across the full customer base, average order frequency is just 1.03, confirming Olist's business is overwhelmingly built on one-time buyers, with little structural mechanism to bring valuable customers back.

Recommended action:

- Launch a targeted win-back campaign for the “Cannot Lose Them” segment before this value is fully lost
- Pilot a post-purchase re-engagement flow (e.g. reminder or incentive at the ~6-month mark) to test whether it improves repeat-purchase rate

Finding 3 — Delivery Delay Predicts Churn

A churn prediction model (Random Forest, ROC-AUC 0.606) was trained on customer purchase behavior, deliberately excluding recency itself to avoid data leakage. Delivery delay emerged as by far the strongest predictor of churn, accounting for 53% of the model's predictive signal — more than five times the influence of total spend or average order value, and far ahead of order frequency. This reinforces Finding 1: shipping inefficiency appears to carry both a margin cost and a customer-retention cost. Model performance is modest in absolute terms and should be read as directional evidence of which factor matters most, not as a production-ready prediction tool.

Recommended action:

- Prioritize delivery-time reliability improvements alongside the freight cost review from Finding 1
- Consider a proactive outreach trigger for customers who experience a significantly delayed delivery, before they go quiet

Methodology Notes

Margin is approximated as (price – freight value) per item, since cost-of-goods data is not available in this dataset; this is a proxy for shipping-cost efficiency rather than true profit. Analysis excludes non-delivered orders (2.98% of all orders) and a small number of products (1.36% of revenue) with untranslated category names. Full methodology, SQL, and underlying notebooks are available in the project repository.